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CO3 – AT2
MLA 0201 – Fundamentals of Machine Learning

Annexure A

Case Study

Case Study 1: Customer Segmentation using K-Means Clustering and PCA A retail store wants to segment its customers based on their Annual Income and Spending Score to design targeted marketing campaigns. Using the dataset below, write a Python program to preprocess the data, apply K-Means clustering, reduce the data to two dimensions using Principal Component Analysis (PCA), visualize the clusters, and identify the optimal number of customer groups.

Dataset (customer_segmentation.csv)

CustomerID	Age	AnnualIncome (k$)	SpendingScore
1	19	15	39
2	21	15	81
3	20	16	6
4	23	16	77
5	31	17	40
6	22	17	76
7	35	18	6
8	23	18	94
9	64	19	3
10	30	19	72

Case Study 2: Dimensionality Reduction using PCA, Factor Analysis, ICA, and Gaussian Mixture Model
A wine manufacturer has collected chemical measurements of different wine samples and wants to reduce the dataset dimensions before performing clustering. Using the dataset below,

write a Python program to standardize the data, apply PCA, Factor Analysis, and Independent Component Analysis (ICA) for dimensionality reduction, then perform clustering using the Gaussian Mixture Model (EM Algorithm) and compare the transformed results.

Dataset (wine_samples.csv)

Sample	Alcohol	MalicAcid	Ash	Alcalinity	Magnesium	Phenols
1	14.23	1.71	2.43	15.6	127	2.80
2	13.20	1.78	2.14	11.2	100	2.65
3	13.16	2.36	2.67	18.6	101	2.80
4	14.37	1.95	2.50	16.8	113	3.85
5	13.24	2.59	2.87	21.0	118	2.80
6	14.20	1.76	2.45	15.2	112	3.27
7	14.39	1.87	2.45	14.6	96	2.50
8	14.06	2.15	2.61	17.6	121	2.60
9	14.83	1.64	2.17	14.0	97	2.80
10	13.86	1.35	2.27	16.0	98	2.98

1: Customer Segmentation using K-Means Clustering and PCA

1. Understanding of the Case

The retail store wants to divide its customers into meaningful groups based on their **Annual Income** and **Spending Score**. This segmentation can help the company design targeted marketing campaigns for customers with similar purchasing behavior.

The important variables are:

- **CustomerID** – unique identifier; not useful for clustering.
- **Age** – demographic information.
- **AnnualIncome** – customer's yearly income in thousands of dollars.

- **SpendingScore** – score representing the customer's spending behavior.

For this case, **AnnualIncome and SpendingScore** are selected because they directly represent the customer's financial capacity and purchasing behavior.

The main objectives are:

1. Preprocess the customer data.
2. Standardize the selected features.
3. Determine the optimal number of clusters.
4. Apply K-Means clustering.
5. Apply PCA for two-dimensional representation.
6. Visualize and interpret the customer segments.
7. Suggest suitable marketing strategies.

2. Application of Concepts

Step 1: Data

Preprocessing

The CustomerID column is removed because it is only an identification

number. The selected features are:

Since Annual Income and Spending Score may have different scales, standardization is performed using:

where:

- = original value
- = mean
- = standard deviation
- = standardized value

This prevents one feature from dominating the clustering process.

3. K-Means Clustering

K-Means divides the customers into groups.

The algorithm works as follows:

1. Select the number of clusters .
2. Initialize cluster centroids.
3. Assign each customer to the nearest centroid.
4. Calculate new centroids.
5. Repeat until the centroids stabilize.

The objective function is:

where:

· = customer data point

· = centroid of cluster

· = number of clusters.

4. Finding the Optimal Number of Clusters

The **Elbow Method** is used.

For different values of , the within-cluster sum of squared errors/inertia is calculated.

Example:

K	Inertia
1	High
2	Lower
3	Lower
4	Lower

5 Lower

The optimal value is selected where the decrease in inertia starts becoming less significant.

For this small dataset, **K = 2** is a reasonable segmentation choice.

In an actual execution, the final K should be selected from the plotted elbow rather than assumed beforehand.

5. PCA

PCA is applied to transform the standardized data into principal components.

The first principal component is:

and the second is:

Since the original dataset already has only two clustering features, PCA mainly provides a convenient transformed coordinate system for visualization.

6. Analysis of Customer Groups

Based on Annual Income and Spending Score, customers can be interpreted into groups such as:

Cluster 0 – Lower/Moderate Income Customers

These customers generally have lower annual income compared with high-income customers. **Marketing strategy:**

- Discounts
- Budget products
- Festival offers
- Cashback
- Student/young-customer promotions

Cluster 1 – Higher Spending/Income Customers

Customers with relatively high income or stronger spending behavior can be targeted with premium offerings.

Marketing strategy:

- Premium products
- Loyalty programs
- Personalized recommendations
- Exclusive offers
- Premium memberships

The exact cluster meaning depends on the cluster centers generated by K-Means.

7. Solution Design

The proposed solution is an automated customer segmentation system: **Customer Data**

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Data Cleaning

↓

Feature Selection

↓

Standardization

↓

Elbow Method

↓

Optimal K

↓

K-Means Clustering

↓

PCA

↓

Cluster Visualization

↓

Targeted Marketing Strategy

This solution is feasible because the retailer can periodically update the customer dataset and rerun the clustering algorithm.

8. Python Program

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler

from sklearn.cluster import KMeans

from sklearn.decomposition import PCA

#

# 1. Load Dataset

#

df = pd.read_csv("customer_segmentation.csv")

print("Original Dataset:")

print(df)

#

# 2. Select Features

#
```

```
X = df[["AnnualIncome", "SpendingScore"]]

#

# 3. Standardization

#
scaler = StandardScaler()

X_scaled =

scaler.fit_transform(X) #

# 4. Elbow Method

#

inertia = []

for k in range(1, 10):

    model = KMeans(

        n_clusters=k,

        random_state=42,

        n_init=10

    )

    model.fit(X_scaled)

    inertia.append(model.inertia_)

plt.figure(figsize=(7, 5))

plt.plot(range(1, 10), inertia, marker="o")

plt.xlabel("Number of Clusters")

plt.ylabel("Inertia")

plt.title("Elbow Method")
plt.show()
```

```
#
# 5. K-Means
#
optimal_k = 2
kmeans = KMeans(
    n_clusters=optimal_k,
    random_state=42,
    n_init=10
)
df["Cluster"] = kmeans.fit_predict(X_scaled)
#
# 6. PCA
#
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
#
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Customer Segmentation using K-Means and PCA")
plt.show()
```

9. Conclusion

K-Means successfully divides customers into groups based on their income and spending behavior. PCA provides a two-dimensional representation that makes the clusters easier to visualize.

The resulting customer segments can be used to create **personalized marketing campaigns**, improving customer engagement and potentially increasing sales.

2: Dimensionality Reduction using PCA, Factor Analysis, ICA and GMM

1. Understanding of the Case

The wine manufacturer has chemical measurements for different wine samples.

The available variables are:

- Alcohol
- Malic Acid
- Ash
- Alcalinity
- Magnesium
- Phenols

The problem is that several chemical variables may contain overlapping information. Therefore, dimensionality reduction is required before clustering.

The objectives are:

1. Standardize the wine measurements.
2. Reduce dimensions using PCA.
3. Reduce dimensions using Factor Analysis.
4. Reduce dimensions using ICA.
5. Apply Gaussian Mixture Model clustering.
6. Compare the transformed representations.
7. Identify a useful representation for clustering.

2. Application of Concepts

Step 1: Standardization

Chemical variables have different units and ranges.

For example:

- Alcohol is approximately around 13–15.
- Magnesium can be around 100–120.
- Other variables have different ranges.

Therefore, standardization is necessary:

After standardization, each feature has approximately:

This prevents large-scale variables from dominating the analysis.

3. PCA

Principal Component Analysis finds directions that capture maximum variance.

The first principal component captures the largest possible variance.

The second principal component captures the next largest variance while being orthogonal to the first.

For example:

The six original features are therefore represented using two principal

components. **Advantage**

PCA is useful for:

- Visualization
- Noise reduction
- Feature compression
- Faster clustering

4. Factor Analysis

Factor Analysis assumes that observed variables are influenced by a smaller number of hidden or **latent factors**.

The general model is:

where:

- · = observed variables
- · = factor loading matrix
- · = latent factors
- · = error/noise

For this case, six chemical measurements can be represented using two underlying factors.

For example, one factor may represent a combination of chemical properties related to alcohol and phenolic characteristics.

5. Independent Component Analysis

ICA attempts to find statistically independent components.

The basic model is:

where:

- · = observed data
- · = mixing matrix
- · = independent components.

ICA differs from PCA because PCA focuses on **variance and orthogonality**, while ICA focuses on **statistical independence**.

6. Gaussian Mixture Model

After dimensionality reduction, GMM is used for clustering.

GMM assumes that the observations come from a mixture of Gaussian probability distributions. The probability density is:

where:

· = number of Gaussian components

· = mixture probability

· = mean

· = covariance matrix.

GMM uses the **Expectation-Maximization (EM) algorithm**.

E-Step

Calculate the probability that each sample belongs to each Gaussian

component. **M-Step**

Update:

- Mean
- Covariance
- Mixing probability

The process continues until convergence.

7. Comparison of Methods

Method	**Main Objective**	**Result**
PCA	Maximum variance	Principal components
Factor Analysis	Latent factors	Factor representation
ICA	Independent signals	Independent components
GMM	Probabilistic clus	

PCA vs Factor Analysis vs ICA

PCA is preferable when the objective is maximum variance preservation.

Factor Analysis is useful when the goal is to discover hidden factors behind observed variables.
ICA is useful when independent underlying sources/components are expected. **GMM** is

useful when clusters may overlap and hard K-Means assignments are not ideal. **8.**

Solution Design

The proposed wine analysis system is:

Wine Chemical Data

↓

Data Cleaning

↓

Feature Selection

↓

Standardization

↓

PCA / Factor Analysis / ICA

↓

2-Dimensional Representations

↓

Gaussian Mixture Model

↓

EM Algorithm

↓

Wine Clusters

↓

Comparison and Interpretation

This approach allows the manufacturer to identify groups of chemically similar wines.
For a larger dataset, the system can be extended to automatically select the number of GMM components using **BIC/AIC**, rather than manually choosing the number of clusters.

9. Python Program

```
importpandasaspd

importmatplotlib.pyplotasplt

fromsklearn.preprocessingimportStandardScaler

from sklearn.decomposition import PCA
```

```
fromsklearn.decompositionimportFactorAnalysis

from sklearn.decomposition import FastICA

fromsklearn.mixtureimportGaussianMixture#

#1.LoadDataset#

df=pd.read_csv("wine_samples.csv") print("Original

Dataset:")

print(df)

#

#2.SelectChemicalFeatures
#

features = [

    "Alcohol",

    "MalicAcid",

    "Ash",

    "Alcalinity",

    "Magnesium",

    "Phenols"

]

X = df[features]

#

# 3. Standardize Data

#

scaler = StandardScaler()
```

```
X_scaled =
scaler.fit_transform(X)
print("\nStandardized Data:")
print(X_scaled)
#
# 4. PCA
#
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
print("\nPCA Result:")
print(X_pca)
print("\nPCA Explained Variance:")
print(pca.explained_variance_ratio_)
print("\nTotal PCA Variance:")
print(sum(pca.explained_variance_ratio_))
#
# 5. Factor Analysis
#
fa = FactorAnalysis(
   n_components=2,
   random_state=42
)
X_fa = fa.fit_transform(X_scaled)
print("\nFactor Analysis Result:")
```

```
print(X_fa)

#

# 6. ICA

#

ica = FastICA(

    n_components=2,

    random_state=42,

    max_iter=1000

)

X_ica = ica.fit_transform(X_scaled)

print("\nICA Result:")

print(X_ica)

#

# 7. GMM Clustering

#

gmm = GaussianMixture(
    n_components=2,

    random_state=42

)

gmm.fit(X_pca)

clusters = gmm.predict(X_pca)

df["Cluster"] = clusters

print("\nGMM Cluster Labels:")

print(clusters)
```

```
#

# 8. Cluster Probabilities

#

probabilities = gmm.predict_proba(X_pca)

print("\nGMM Cluster Probabilities:")

print(probabilities)

#

# 9. Comparison Table
#

 comparison = pd.DataFrame({

        "PCA_1": X_pca[:, 0],

    "PCA_2": X_pca[:, 1],

    "FA_1": X_fa[:, 0],

    "FA_2": X_fa[:, 1],

    "ICA_1": X_ica[:, 0],

    "ICA_2": X_ica[:, 1],

    "GMM_Cluster":

 clusters })

print("\nComparison:")

print(comparison)

#

# 10. PCA + GMM Visualization

#

plt.figure(figsize=(8, 6))
```

```
plt.scatter(

    X_pca[:, 0],

    X_pca[:, 1],

    c=clusters,
    s=100

)

plt.xlabel("PCA Component 1")

plt.ylabel("PCA Component 2")

plt.title("PCA + Gaussian Mixture Model")

plt.show()

#

# 11. Factor Analysis

Visualization #

plt.figure(figsize=(8, 6))

plt.scatter(

    X_fa[:, 0],

    X_fa[:, 1],

    s=100

)

plt.xlabel("Factor 1")

plt.ylabel("Factor 2")

plt.title("Factor Analysis")
```

plt.show()

#

#12.ICAVisualization#

pl~~t.figure(figsize=(8,6))~~

plt.scatter(

X_ica[:,0],

X_ica[:,1],

s=100

)

```
plt.xlabel("IndependentComponent1")
plt.ylabel("Independent Component 2")
plt.title("IndependentComponentAnalysis")plt.show(
)
```

10. Analysis and Interpretation

The three dimensionality-reduction techniques provide different representations of the same wine samples.

PCA

PCA identifies directions with maximum variance. The explained variance ratio indicates how much of the original information is retained by the two components.

Factor Analysis

Factor Analysis represents the observations using underlying latent factors. It is useful when several chemical properties are assumed to arise from common underlying characteristics.